Solution The given quadratic equation is, $x^2(p^2+q^2)+2x(pr+qs)+r^2+s^2=0$

Here, $a=p^2+q^2,\ b=2(pr+qs),\ c=r^2+s^2$

Now, $\Delta=b^2-4ac$ $=\left[2(pr+qs)\right]^2-4(p^2+q^2)(r^2+s^2)$

$=4\left[p^2r^2+2pqrs+q^2s^2-p^2r^2-p^2s^2-q^2r^2-q^2s^2\right]$

$=4\left[-p^2s^2+2pqrs-q^2r^2\right]=-4\left[(ps-qr)^2\right]<0$ ...(1)

since, $\Delta=b^2-4ac<0$, the roots are not real.

If $ps=qr$ then $\Delta=-4\left[ps-qr\right]^2=-4\left[qr-qr\right]^2=0$ (using (1))

Thus, $\Delta=0$ if $ps=qr$ and so the roots will be real and equal.

Exercise 3.13

1. Determine the nature of the roots for the following quadratic equations

   (i) $15x^2+11x+2=0$ (ii) $x^2-x-1=0$ (iii) $\sqrt{2}t^2-3t+3\sqrt{2}=0$

   (iv) $9y^2-6\sqrt{2}y+2=0$ (v) $9a^2b^2x^2-24abcdx+16c^2d^2=0,\ a\neq0,\ b\neq0$

2. Find the value(s) of 'k' for which the roots of the following equations are real and equal. (i) $(5k-6)x^2+2kx+1=0$ (ii) $kx^2+(6k+2)x+16=0$

3. If the roots of $(a-b)x^2+(b-c)x+(c-a)=0$ are real and equal, then prove that b, a, c are in arithmetic progression.

4. If a, b are real then show that the roots of the equation $(a-b)x^2-6(a+b)x-9(a-b)=0$ are real and unequal.

5. If the roots of the equation $(c^2-ab)x^2-2(a^2-bc)x+b^2-ac=0$ are real and equal prove that either $a=0$ (or) $a^3+b^3+c^3=3abc$

Thinking Corner

Fill in the empty box in each of the given expression so that the resulting quadratic polynomial becomes a perfect square.

(i) $x^2+14x+\square$ (ii) $x^2-24x+\square$ (iii) $p^2+2qp+\square$

3.6.7 The Relation between Roots and Coefficients of a Quadratic Equation

Let α and β are the roots of the equation $ax^2+bx+c=0$ then,

$$\alpha=\frac{-b+\sqrt{b^2-4ac}}{2a},\quad \beta=\frac{-b-\sqrt{b^2-4ac}}{2a}$$

From 3.6.3, we get

$$\alpha+\beta=\frac{-b}{a}=\frac{-\text{Co-efficient of } x}{\text{Co-efficient of } x^2}$$

$$\alpha\beta=\frac{c}{a}=\frac{\text{Constant term}}{\text{Co-efficient of } x^2}$$

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